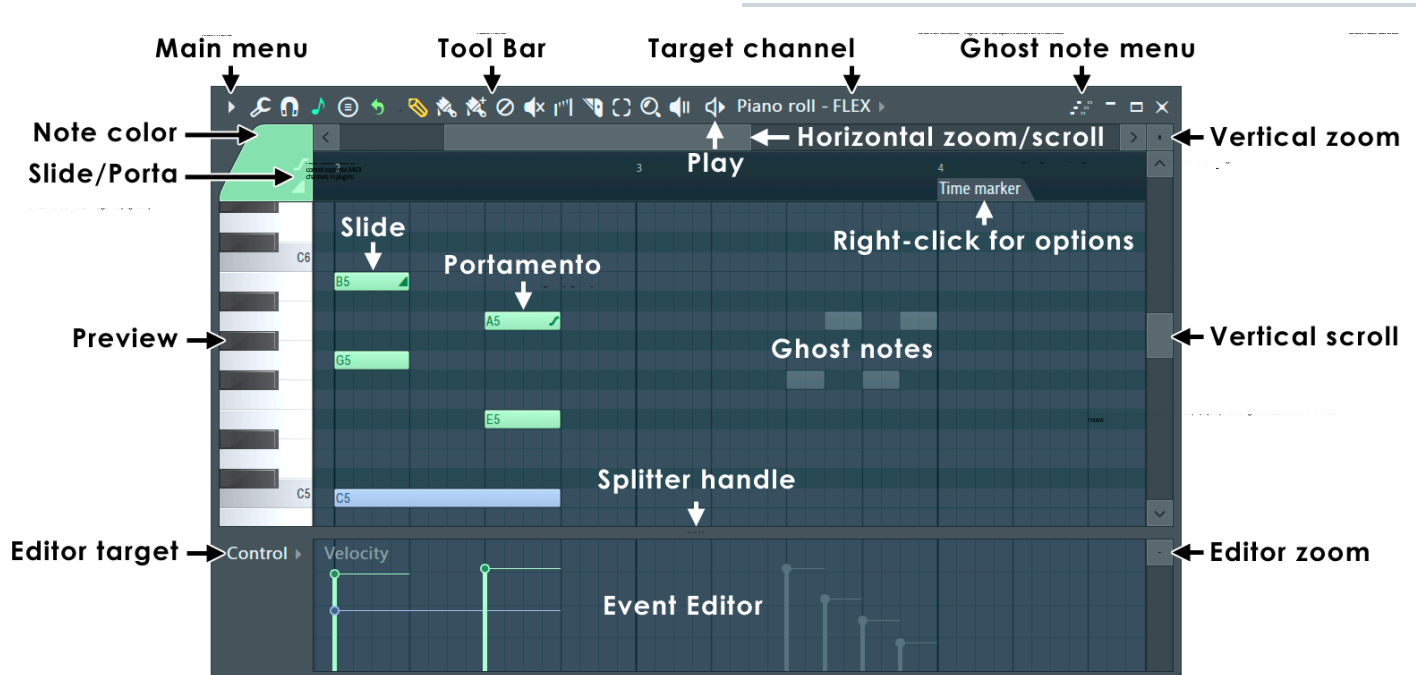


FL-Studio PIANO ROLL Manual

Piano roll

FL Studio's **Piano roll** has the well deserved reputation as the best Piano roll in the business. The Piano roll sequencing process is known as 'sequencing'. There are a number of useful tools to help with complex score editing and on the horizontal axis. This is the exact principle as paper '[Piano rolls](#)' used to automate player-pianos in former times of unlimited complexity. Note data can be entered manually with the editing tools or recorded in from 'live' [MIDI](#)



NOTES: You can click the **(Middle-Mouse-Button)** and drag in the Piano roll to scroll vertically and horizontally. You can also right-click a note to change its custom color, **(Right-click)** it.

Piano roll Basics

- **Notes** - are displayed as horizontal bars. **Slides** are shown as horizontal bars with a small triangle drawn on them.
- **Pitch** - is mapped from bottom (low) to top (high).
- **Duration** - The horizontal length represents time. The number on the Piano roll's top-ruler are bars (gr).
- **Resolution / grid** - The smaller grid each bar is divided into are set by the window's [snap](#) () setting. **with Zoom** when Snap is set to **Line**. Selecting **Main** will use the [Global Snap](#) value. **Some Global snap**
- **Target Channel** - Select a Channel with the drop-down menu, in the title bar, to select for editing. Channels are shown while hovering over the control to show Channels with notes **OR** hold **(Ctrl+Mouse-wheel)** to show us
 - **Auto locate** - When Patterns are changed the Piano roll will select the first Channel with notes.
- **Opening from the Playlist** - If the **Playlist Menu** option '[Edit patterns in > Piano roll](#)' is selected, when

NOTES:

1. **Click-and-hold** - Beware the '[F10 > General Settings > Click-and-hold functions in piano roll](#)' option of
 - **Copy note**: Hold on an existing note to copy it.
 - **Cycle slide/porta** - Click and hold when adding a note to cycle through slide & porta event modes.
 - **Glue notes** - Place the cursor between 2 neighbor notes, so the resize cursor appears, then click.
2. **Mouse-wheel velocity** - Hold (**Alt/Opt**) and use Mouse-wheel while hovering notes to change velocity.
3. **Mouse-wheel tools** - Change tools by holding right click and rolling the wheel.
4. **Time Markers** - To add make sure you are in **Pattern Mode** and use to add a Time Marker where the **length (Right+Click)** the Time Marker to define the length of the currently selected Pattern.

Piano roll Controls

- **Toolbar icons** - Along the top of the Piano roll window:



[Main Menu](#) - Includes: Edit, Tools, View, Snap, Select, Group, Zoom, Time Marker, Clip S



[Tools menu](#) - Riff machine (**Alt/Opt+E**); Generate chord progression (**Alt/Opt+P**); Quick (**Shift+Q**); Quantize (**Alt/Opt+Q**); Quick chop (**Ctrl+U**); Chop (**Alt/Opt+U**); Glue (**Ctrl+G**); (**Alt/Opt+K**); Flip (**Alt/Opt+Y**); Randomize (**Alt/Opt+R**); Scale levels (**Alt/Opt+X**), LFO (**A**)



[Piano roll Snap](#) - Snap determines how Notes will move and quantization aligns events. a **Global Snap** selector on the [Snap Panel](#).



Snap to scale - (**Left-Click**) to activate and deactivate the **Snap to scale** - Notes are sn to select a **Root note** and **Scale** plus set **Snap incoming MIDI** - So your controller Key



Stamp tool - The Chord Tool lets you pick a Chord and automatically draws the correct builds it for you according to the selected scale (see **Snap to scale** above). Choose a ch you click in. **NOTE:** Deselect '**Only one**' at the top of the menu to avoid the tool reverti

Automatic chord - This special mode deserves some more explanation; When '**Only on** the [Chord Progression Tool](#) - without the need to interact with the interface. Just set th will be created for you.

How it works - The 'Automatic chord' mode considers the Scale from Snap to scale set chords on either side of the location where a Chord is added. The Chord will be inverted time interval of an existing chord will regenerate that Chord. If no scale is specified, the

existing non-chord notes (such as high melody or low bass notes), this tool can be used. If you want, try stamping over the existing ones a few times until you get a progression you like.



Loop switch - See the Channel Rack section on [Looping Controls](#). In this case the Piano Roll will jump to the current loop point.



Draw (P) - (**Left-click**) to add a note. (**Left-click-and-drag**) to reposition the note before it is added. (**Click and hold**) on the background to open the slice tool (F10 > General settings > Click and hold functions on). (**Alt**) to add in Draw mode (pencil) with snap set to 'none'.



Paint (B) - Paints multiple notes as you drag in lines. You can paint notes above or below existing notes. **Right-Click notes** to delete them. **Modifier keys: (Shift)** - **Monophonic paint** all notes to a single pitch. **Monophonic** score. Use (**Ctrl**) to make a selection.



Paint drum sequencer mode (N) - Similar to the 'Paint' tool above. Adds multiple notes in a drum pattern where you may want to mute/unmute steps. **Left-click** to add a note. Click-and-drag to add multiple notes. **Right-Click and drag** to delete multiple notes. **Snap** sets the note length. (**Right-Click**) to toggle functions on). **Modifier keys: (Ctrl)** to make a selection. (**Alt**) paint at half the selected step length.



Delete (D) - Click or Click-and-drag to delete notes. (**Right-Click and hold**) on the background to open the delete tool (F10 > General settings > Click and hold functions on). **Modifier keys: (Ctrl)** to make a selection, then press (**Delete**).



Mute (T) - (**Left-click**) on notes to mute them. (**Left-Click and drag**) to mute multiple notes. (**Click and hold**) on the background to open the mute tool (F10 > General settings > Click and hold functions on). **Modifier keys: (Ctrl)** to make a selection.



Interpolate (I) - This special tool only functions when actual [Event Data](#) is shown in the Event Editor. This transforms stepped data into straight-line gradients, creating a smoother progression. It does not apply to 'lollipop-style' note properties.



Slice (C) - (**Left-click**) and drag vertically to slice through notes. **Glue notes** by making a selection and dragging. (**Ctrl**) to make a selection. (**Shift+Alt**) Slice without dragging and with snap set to 'none'.



Select (E) - **Click** notes or **Click and drag** to make group selections. **Modifier keys:** (Sh



Zoom to selection (Z) - (**Left-Click and drag**) to zoom to selection. (**Left-click on back** selection. (**Right-Click**) again to swap between zoom and unzoomed modes. (**Right-Click**

- **Zoom in/ Zoom out (Page up / Page down)** - Zoom centers on the Cur
- **Quick-zoom 1, 2 & 3 (Shift+1, 2, 3)** - Three handy zoom levels.
- **Zoom out far (Shift+4)** - Minimum horizontal zoom.
- **On Selection (Shift+5)** - Zoom to time-line selection. The command will b

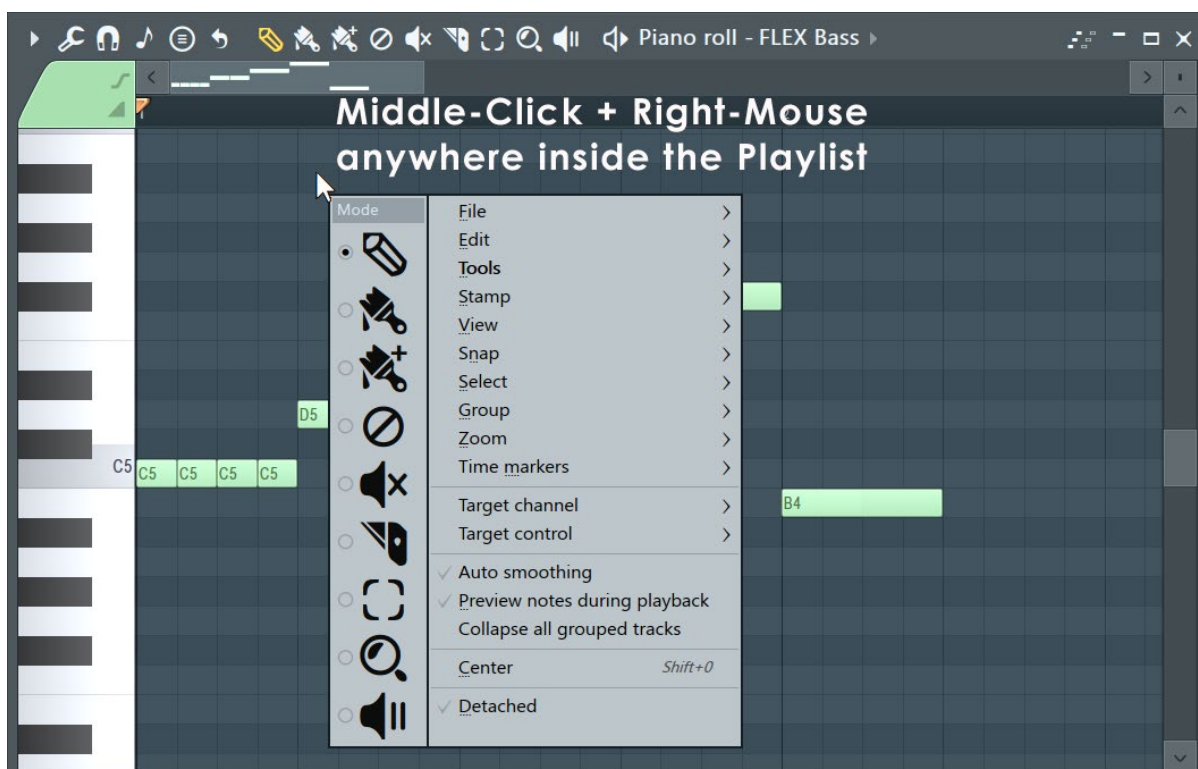


Play selected (Y) - **Click** the notes you want to play. **Click and drag** to 'scrub' play note



Play / Pause - **Left-click** to start/pauses playback. **Right-Click (while playing)** to stop a start of the Piano roll. **NOTE:** There is an option on the (**Right-Click**) main transport Sto

- **Piano roll Context Tool-Menu** - The Context Tool Menu provides local access to the above Icon contr open it, use (**Middle-Mouse-Click+Right-Click**) anywhere inside the **Piano roll**, including on existing



- **Target Channel** - Change instrument channels (of the current pattern) without leaving the Piano roll. (F
- **Event editor target** - Selects the data to be displayed in the integrated event editor. This can include p

- **Slide & Portamento** - These are flags that can be set per-note and work as follows:
 - **Portamento Toggle** ( O) - When selected, notes added to the Piano roll will display a 'portamento' flag on **the note** with the **portamento flag**. This is a note transition effect. **NOTES:** 1. these features only work in **piano roll** - Click and hold when adding a note to cycle through slide, portamento note mode.
 - **Slide Toggle** ( S) - When selected, notes added to the Piano roll become '**slide notes**' displaying a slide-note. The slide-time is set by the length of the slide-note.
- **Note color / Color group selector** - Create note groups for independent editing of overlapping notes
 - **MIDI Control** - Color groups appear to plugins as [separate MIDI channels](#), so one Piano roll can have multiple channels.
 - **Recording notes** - To record incoming MIDI Channels from a controller or MIDI sequencer to the Piano roll, use the [channel through](#) option.
 - **Color group selection** - There is another **Note / Clip Grouping** function activated from the [Tools menu](#).
 - **Customize note colors** - Right-click the Note Color selector to change note colors from the default.
- **Preview keyboard** - Click on the keyboard to preview notes. This does not work when play is active. Click on the note you click.
- **Horizontal & Vertical Zoom controls** - To horizontally zoom the Piano roll click on the edge of the horizontal scroll bar. To vertically zoom, click on the edge of the vertical scroll bar.
- **Ghost notes** - These are notes in other Piano rolls associated with the current Pattern. See the [Piano roll menu](#) (or **X1 button**) on the notes to switch to the Ghost channel. **Piano roll menu > Editable ghosts (Ctrl+G)**
- **Event Editor** - Edit: Note Velocity, Pan, Pitch, Filter cutoff & [Automation events](#) (the lower Piano roll area). Right-click the area at the front of the editor as shown above.
- **Notes** - Click/select to drag, stretch, paint, clone, copy, etc. (see 'operating with notes' below). **Note color selector**
- **Splitter handle** - Left-click and drag the divider to resize the upper and lower windows. Right-Click to lock/unlock.

Using the Piano roll

FL Studio has, arguably, the most powerful Piano roll editor available. Combined with the [Piano roll menu](#) you can do a lot.

NOTE: Many of the movement and draw commands are constrained by the [snap setting](#)  that controls how notes snap to the grid. Also, the [Global snap setting](#) can be used for **MIDI input quantizing**.

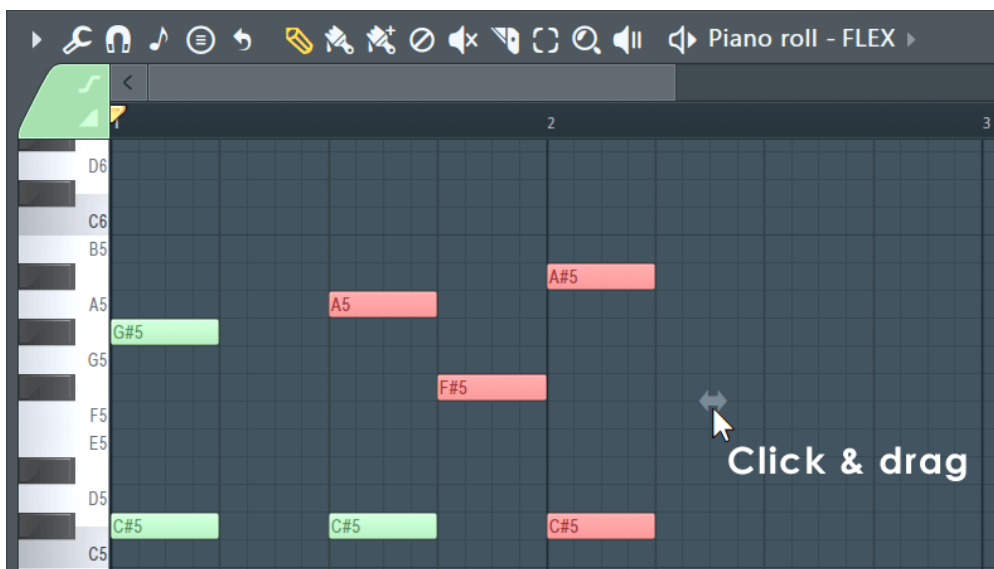
- **Tools Menu** - () is a shortcut to the Piano roll tools submenu (see the [Piano roll menu](#) page, ) , click on the Tools menu.
- **Adding Notes - draw mode** () allows you to draw, edit and delete notes. To **add notes and draw** notes, click on the **draw mode** button. **Draw mode** is similar, but allows you to draw multiple notes at once while dragging horizontally in the Piano roll. **Note color selector** - In draw or paint mode, **Left-click** in the Piano roll's grid to **draw a note**. The note will be added to the current channel.
- **Deleting notes** - **Right-Click a note to erase** it (alternatively switch to **erase mode** to erase with Left-click).
- **Selecting/Deselecting groups of notes** - There are a number of methods to select several notes, so you can edit them together.

- **Standard selection** - Press and hold (**Ctrl key**) and either **click a note** to select it, or **drag a rectangle** to select multiple notes.
- **Magic lasso** - The **Piano roll Menu > Select > Magic lasso** option must be selected. Move your mouse (remember the 'magic' remember). **Make sure** you have no notes selected before you perform the lasso action.
- **Add notes to selection** - Hold (**Ctrl+Shift**) together while selecting to add notes to the existing selection.
- **Selecting notes in a time or pitch range** - **Time**, (**Ctrl+Left-click**) or double-click the time ruler. **Pitch**, (**Ctrl+Left-Click**) the **Preview Keyboard** to select a range of notes with the same pitch (dragging).
- **Deselect all notes** - While holding (**Ctrl key**), click empty space in Piano roll's grid.
- **Remove notes from selection** - (**Ctrl+Shift**) deselects individual notes from the existing selection.
- **Invert selection** - Use (**Shift+I**).
- **Select mode button** - Instead of holding (**Ctrl key**), you can also switch Piano roll to **select mode**.
- **Group notes** - (**Shift+G**) - Group the selected notes. Grouping functions so that an action on any note affects the group. **To group notes:** make a selection using the  Select tool, then use the Group function. **Action:** Deselecting the Note / Pattern Grouping switch will disable note grouping behavior until the switch is re-enabled.
- **Copy & Paste** - Make a selection (see above), hold the (**Shift**) button THEN drag selection with the left mouse button.
- **Duplicate selection** - Make a selection and press (**Ctrl+B**). The selection, or all notes if nothing is selected, will be duplicated. **NOTE:** The time-line selection will be duplicated, the notes duplicated by the note selection. In this way, notes outside the time-line selection will be duplicated.
- **Moving & resizing notes** - Select the note/sequence, **Left-click on the note** and drag vertically or horizontally. A double-headed arrow when you can click and drag to resize. Some tips...
 - **Snap setting**  - This will affect horizontal the movement, causing notes to jump between snap points.
 - **Change note length/s** - You can change the length of single notes or selections of notes. **NOTE:**
 - **Resizing notes from the end** - Move the cursor to the right edge of the note and when you click, the note will be resized.
 - **Resizing notes from the start** - Move the cursor to the left edge of the note and when you click, the note will be resized. If the left is not enabled by default, select the '**Piano roll menu > Edit > Allow resizing from left**'.
 - **Horizontal & vertical lock** - Often when moving notes you want to preserve their vertical position.
 - **Vertical position lock (pitch)** (**Shift**) - **NOTE:** First click the note or selection THEN hold the **Shift** key.
 - **Horizontal position lock (timing)** (**Ctrl**) - **NOTE:** First click the note or selection THEN hold the **Ctrl** key.
 - **Nudging notes & selections** - This allows you to move the note/s by either 'snap' or 'pixel' based on your keyboard. The snap setting  will affect the bump size. **Pixel units:** Hold the (**Alt**) key and move the mouse.
 - **Mouse Wheel** - Horizontal note positions can be adjusted by (**Shift+Alt/Opt+Mouse-Wheel**) by moving the mouse wheel. **NOTE:** In the **Edit > Nudge notes with the mouse-wheel** that will allow you to use the Shift alone with the mouse wheel.

- **Edit Ghost notes** - Use (**Double Right-Click** or **X1 button**) on the Ghost notes to switch to the Ghost
- **Stretch / compress scores** - Select a group of notes and drag the [pop-up handle](#) OR hold (**Right-Shift**) duration. (**Alt + Right-Shift**) will stretch in 25% increments.
- **Slice Tool** - () allows you to split one or more notes in the direction of the cut through the Piano roll area and drag to define the "cut line" direction and length. Release the mouse button to split all notes
- **Resize chords & splits** - Hold (**Left-Shift**) click and drag on the slice point or right-side of the chord.
- **Playback/Scrub tool** - () enables you to preview the current sequence by dragging horizontally in the **Right-Click** while in **draw mode**.
- **Play notes straddling the playback position marker** - When stopping and starting you can play notes
- **Quick chords** - Enter chords with one click using the [Piano roll Menu > Tools > Stamp Tool](#) . The s
- **Note properties** - **Note velocity, panning, filter-cutoff**, etc, are available from the **Target control** me **wheel** (**Alt/Opt+mouse-wheel**) by holding the cursor over a target note in the 'Note event window'.
- **Merging scores across Patterns** - Place the Pattern Clips in the [Playlist](#), with the Pattern Clip you want **pattern Clips** function from the [Playlist Menu](#). This merges the selected Pattern Clips into the uppermo
- **Click-and-hold functions in piano roll** - **Copy**: Hold on an existing note. **Slide & Porta**: Click and hold

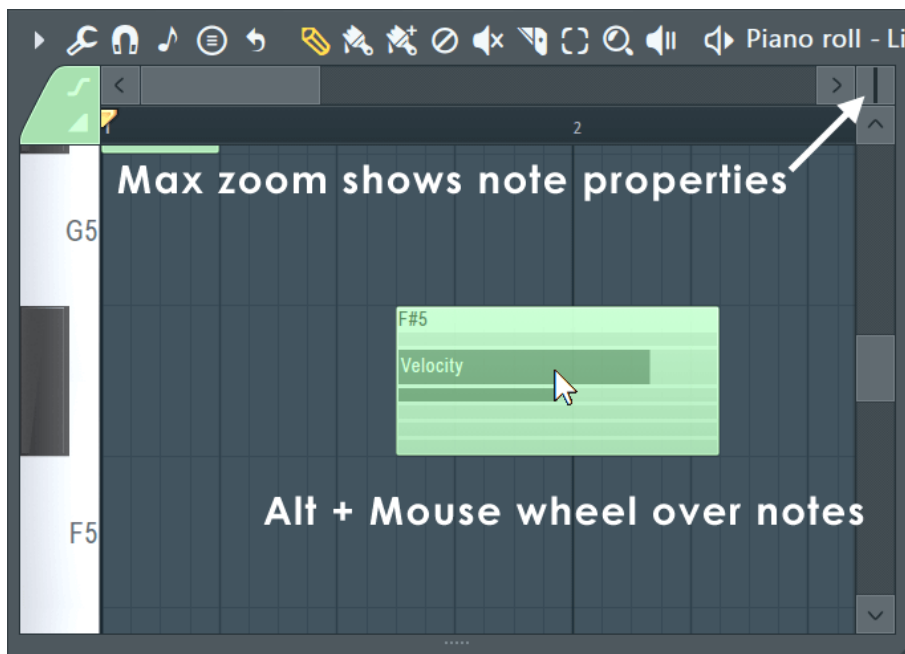
Selection Stretch Handle

After making a selection the **Stretch Handle** appears. You can click and drag this to stretch the selected notes right-side of a note in the group and drag.



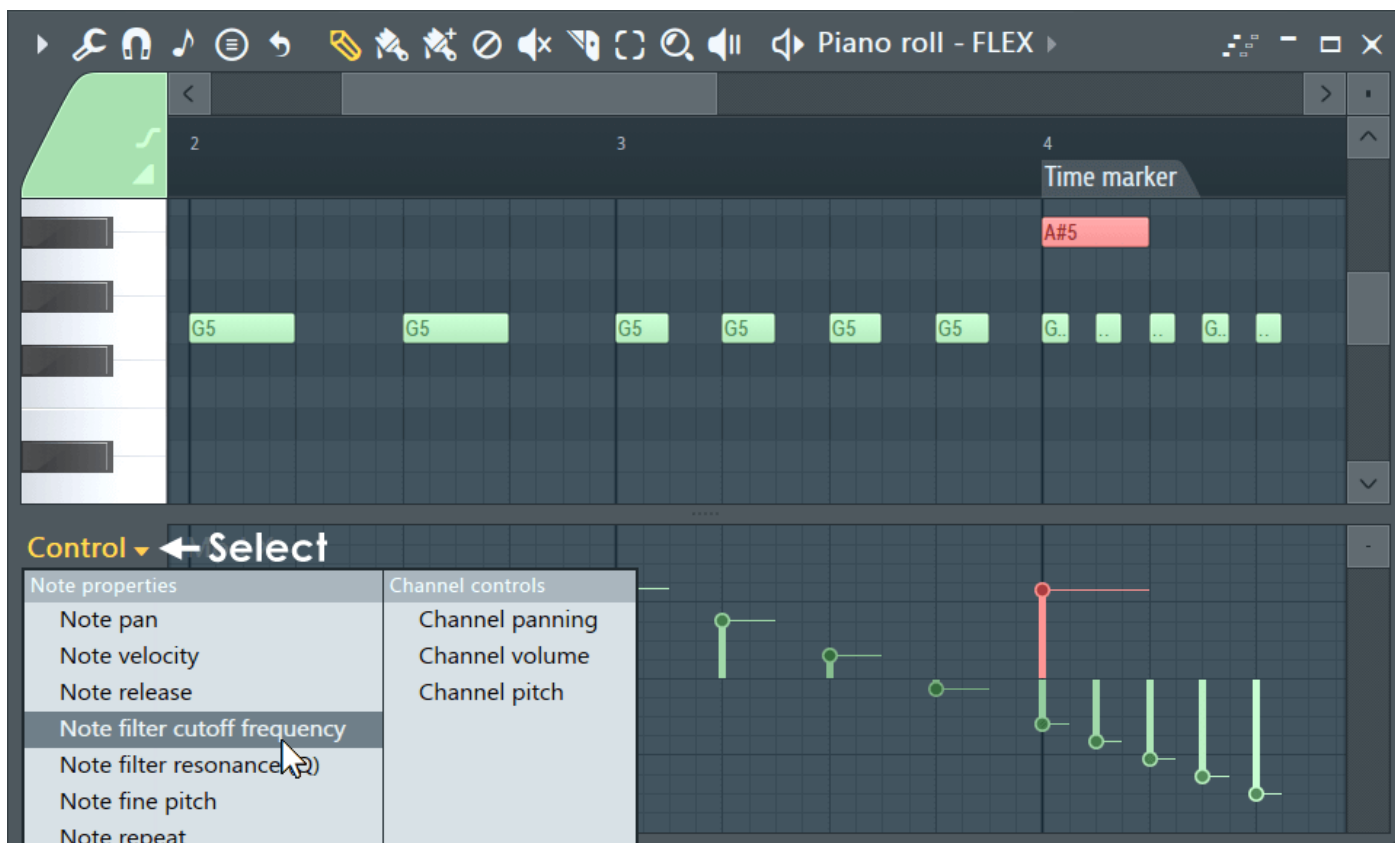
Mouse-wheel Property Note Changes

Hold your mouse over the note and hold (**Alt/Opt+Mouse-wheel**) to change the currently selected note property (**Ctrl+Alt/Opt+Mouse-wheel**)



Note Properties & Event Editor

The **note properties editor** gives you access to a number of per-note properties and event data velocity, pan, and the **Target control**, as shown below.

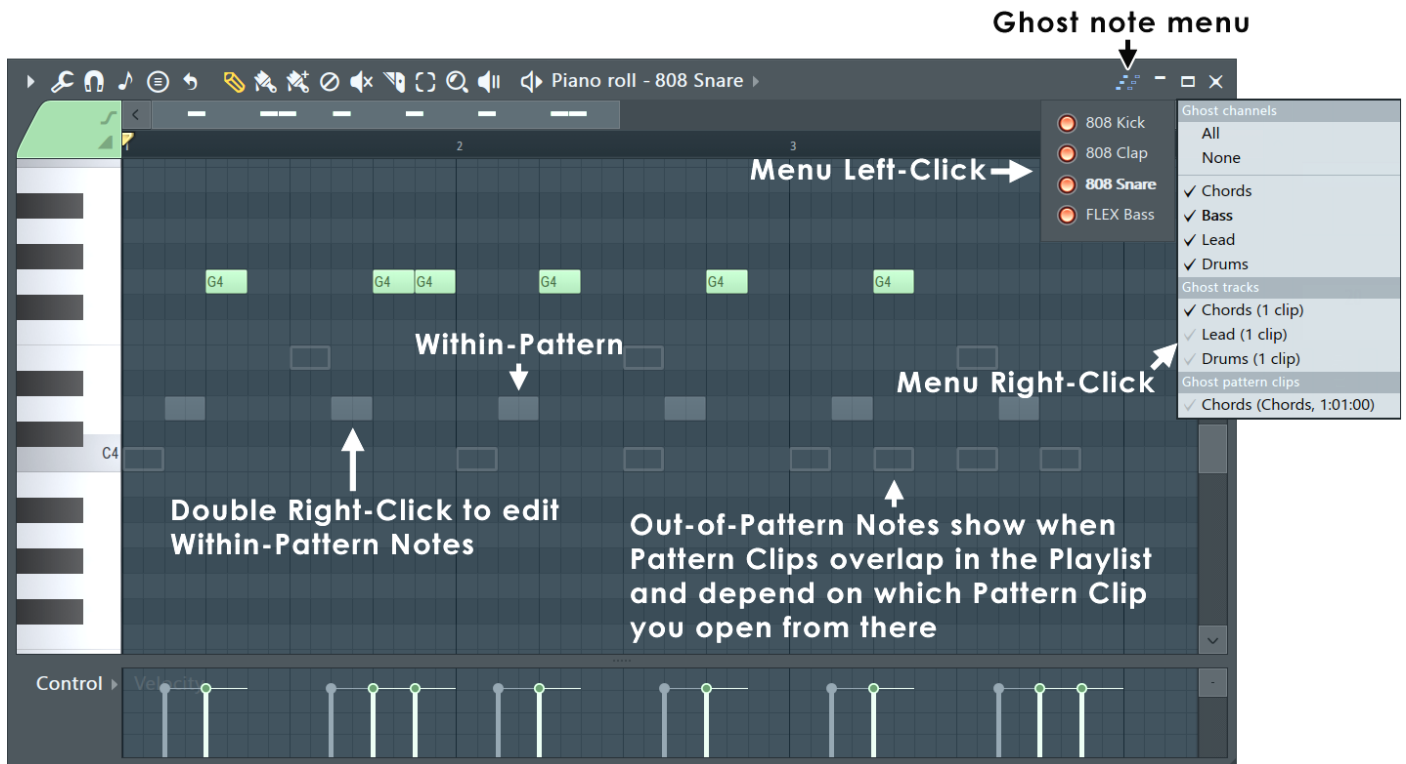


- **Selecting Targets** - Left-click to select from the menu. You can also roll your (**mouse-wheel**) while hovering over the notes.
- **Note properties** - are displayed as vertical lines with a small circle at the top. The right-facing tail allows you to select the notes they belong to. The editor can also display standard [Event Automation](#) associated with the property. To set each note's property to the same value, first, then edit the property for the note (as shown above in red). To set each note's property to the same value, use (**Alt/Opt+Left-Click**).

- **Note events** - When editing Automation Events the editor works in the same way as the [Event Editor](#).

Ghost Notes

Ghost Notes show note data from other Channels (Piano rolls). They are a great help when creating melodies. Those within the same Pattern (Solid blocks) and those from other Patterns (Open Blocks). See below to learn



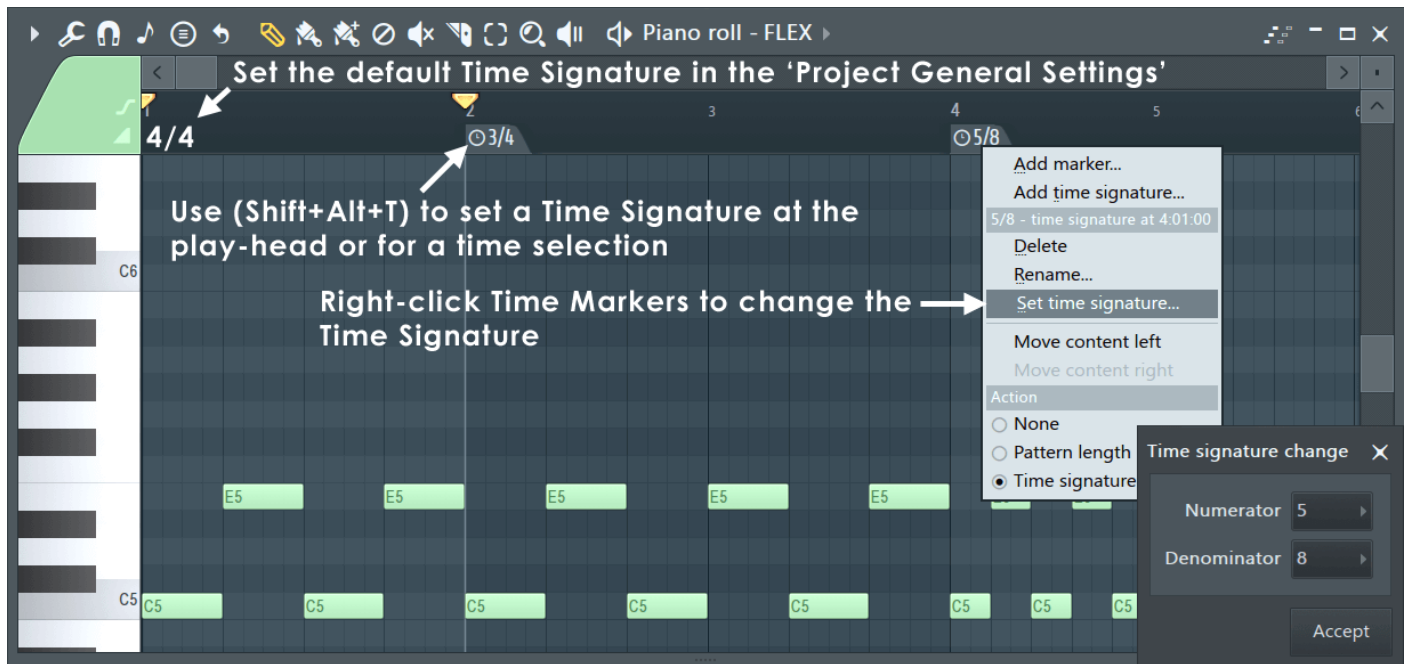
NOTE: When working with overlapping **Patterns in the Playlist**, it matters which **Pattern Clip** you open to edit. You are working on, and so which out-of-pattern Ghost Notes to show.

- **Ghost Note types:**
 - **Within Pattern Ghost Notes** (Solid blocks) - You can (**Double-Right-Click**) to switch to the Channel.
 - **Out of Pattern Ghost Notes** (Open blocks) - These Notes show when Pattern Clips overlap in the Playlist. These Notes move through the Piano roll.
- **Ghost Note Menu (Icon)** - Located on the Piano roll Toolbar can be Left or Right Clicked ...
 - **Left-Click Panel** - Click to enable or disable the Ghost Note Channel selector. When selected, the notes display as Ghost Notes.
 - **Right-Click Menu** - This menu replicates some features of the Panel but includes additional options to deactivate which Playlist Tracks display Ghost Notes (**Ghost tracks**). **Note:** This duplicates the (Pattern) individual Pattern Clip instances (**Ghost pattern clips**).

Time Signatures

The default Project Time signature is set in the '[Project General Settings > Time settings > Set as time signature](#)' independent of the [Playlist Time Signature](#), you can use multiple Time Signatures at any point in the Playlist (piano roll).

to **manage your Pattern and Playlist time signatures independently.**



NOTE: Time Signatures are a visual helper for the Piano roll and Playlist grids. They have no technical effect on notes. You can change Time Signatures for the Pattern without affecting the notes, only the grid behind them changes. **One** Time Signature change in the Piano roll (Channel) will affect **ALL Channels** in the Pattern. Use the Pattern Clip menu option '**Make unique**' to change the Time Signature for a single channel.

TIPS: Here are some useful tips for working with Time Signatures. This applies both to the Piano roll and Playlist.

- **Set a custom Time Signature** - Place the Play head at the point you want to set the new time signature.
- **Set a Time Signature range** - Make a time selection and hold (**Shift+Alt/Opt+T**) to open the Time Signature menu.
- **Set a Time Signature based on the Pattern Clip location in the Playlist** - Open the [Clip Menu](#) and select the Time Signature.
- **Move Time Signatures** - Drag the Time Marker/s. Additionally you can use this feature to set partial bars.
- **More on Time Signatures** - See: [Playlist Time Signatures](#) & [Project General Settings](#).

Scales & Keys (Key Signatures)

While there are always exceptions to this rule, if you **use notes within a Key Signature** your melodies and, to a large extent, your harmonies, will be much more coherent. This is such a powerful tool for composing, particularly if you are new to music and don't have a good grasp of what a scale is. After reading this section. Let's dig a little deeper into this topic ...

In **Western music**, there are seven 'natural' notes named — A, B, C, D, E, F and G. These are the white notes on a piano keyboard. When you add 'sharp' or 'flat' intervals, the total number of notes rises to 12. These are the black notes on a Piano keyboard. **From the tonic** (starting, or **tonic** note, is repeated at the beginning and end of the scale. More specifically, the **important factor** in defining a scale is its tonic (starting) note. It is the relationship between notes, their pitch intervals that defines the scale, not the absolute pitch of the notes or the semitone intervals from note-to-note. There are two very common scales - Major and Minor.

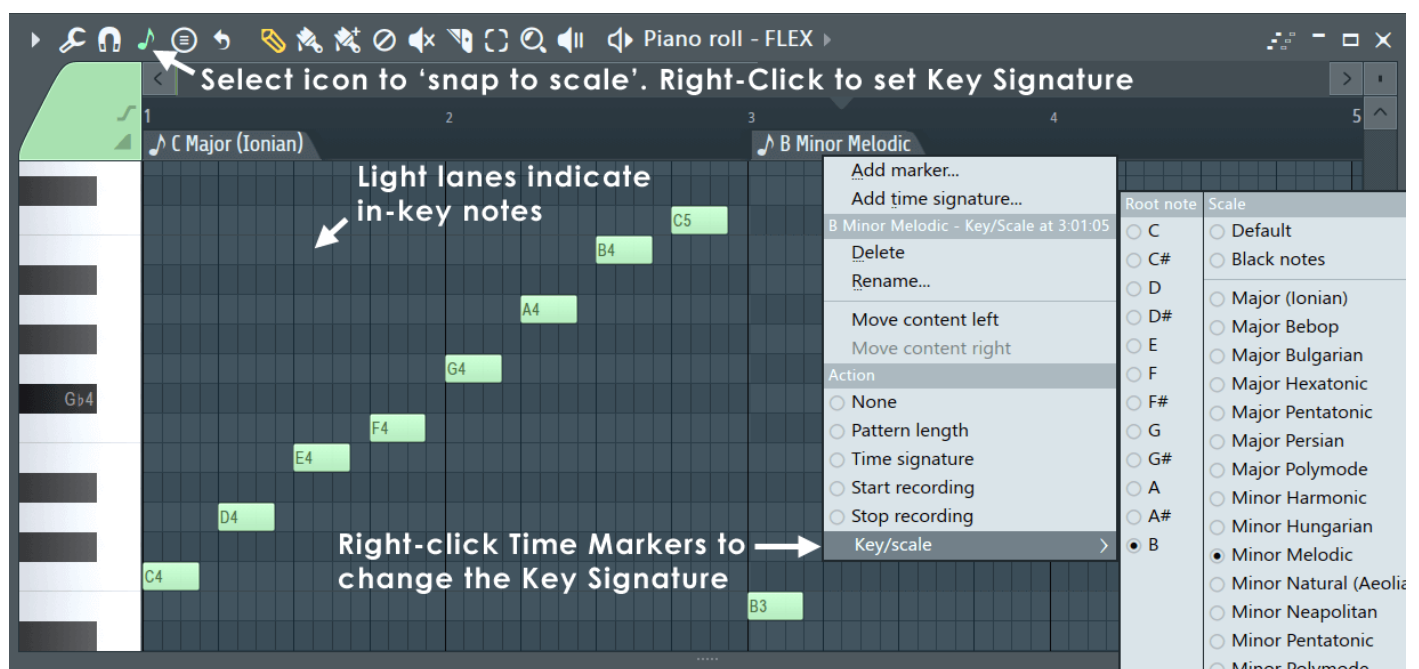
All ascending **Major scales** have the following sequence of pitch intervals from one note to the next - Tone (2 semitones, 200 cents), Semitone (100 cents). This allows you to start on any note and create a Major scale by following this sequence of intervals, regardless of the note you start on. If you check the '**Scale**' list shown in the Scale menu below, these are just the first few scales in the list.

Importantly, by using notes only from within a given scale, you can achieve harmonious sounds, making scales (and note intervals).

The most **common scales** used in music are the **Major (Ionian)** and **Minor (Aeolian)** scales. Starting at the tonic, in a **Major scale** this pitch interval is a **Major third**, four semitones. In a **Minor scale**, the interval is a **Minor third**, three semitones. The Major third in Major scales creates a brighter, more positive sound. The Minor third in Minor scales has a darker, more somber sound. Music is about feelings. Particularly Jazz which creates feelings of confusion and anger. Listen to those who claim superiority because they understand it.

TLDR:

The **Key** or notes you should use, are defined by two things - **Tonic** (starting note) and **Scale** (note intervals). You generally start and end your melodies and bass lines, on or near this tonic note, as this will reinforce the Key. You didn't need to read any of the four paragraphs above! Just start clicking around and enter notes that sound good.



NOTES: The Piano keyboard is designed around the C Major Scale. If you start on C and play only the white notes, you are playing the C Major Scale. When you combine a **Tonic** (starting note) and **Scale** this defines a **Key** for the music (a '**Key Signature**'). The **Key Signature** is the key for the music. Of course, sometimes playing notes out-of-scale is also cool.

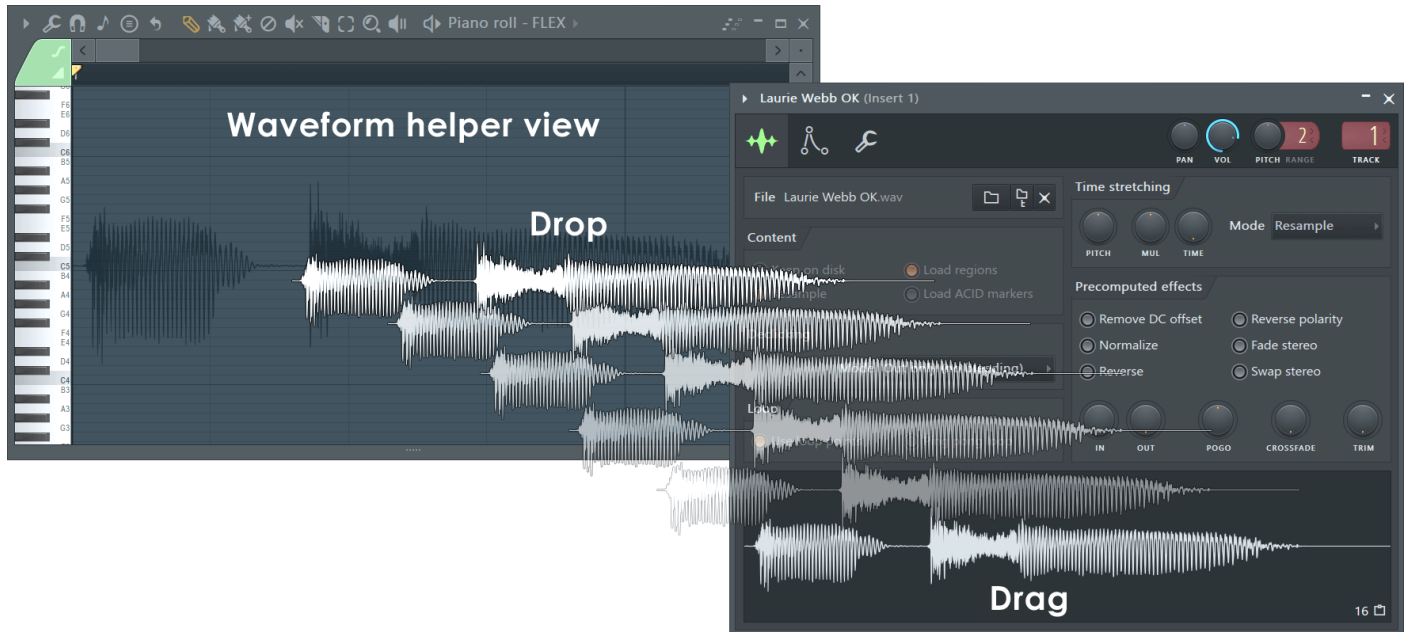
TIP: (Right-click) the **Snap to scale** (note icon) and select either the **Major (Ionian)** or **Minor (Aeolian)** scale. Scales, Keys and Chords are the building blocks of popular music.

Settings

- **Note (icon)** - From the Piano roll Toolbar:
 - **Snap to scale** (Left-Click) - When selected, dragged notes will snap to the Scale and Key (highlighted). If you don't select a scale, they fall out-of-scale after settings are changed.
 - **Select a scale** (Right-Click) - Choose a Tonic, or first note, (left side) and Scale (right side) on the piano roll. The selected scale is highlighted in a lighter color. A **Key Signature** includes a Tonic note and Scale. Right-click the note icon to set the Key Signature.

Waveform Helper View

Sometimes it is useful to closely align notes with an audio waveform. For example when using Fruity Vocoder or Pitcher. To show the waveform drag from a waveform preview and drop on the Piano roll. To **deselect** this view use **Piano roll menu > View > Background waveform (Alt/Opt+N)**.



Understanding Slides & Portamento

The following **ONLY applies to native FL Studio instruments**. To slide a VST instrument automate the [channel pitch](#) knob. The Piano roll can slide notes from one pitch to another using **slide notes** above or below the notes to be affected. Slide notes have a small triangle drawn at the start of the note.

- **Portamento Toggle** () - When selected, notes added to the Piano roll will display a 'portamento flag' ( icon). The portamento flag causes a short slide in pitch from the end of one note **to the note** with the **portamento flag**. This is a note transition effect. **NOTES:** 1. these features only work with native FL Studio instruments and not VST instruments. 2. **Click-and-hold functions in piano roll** - Click and hold when adding a note to cycle through slide, portamento note mode. See [General Options](#) settings.
- **Slide Toggle** () - When selected, notes added to the Piano roll become 'slide notes' displaying the slide icon () that causes notes, above or below the slide-note, to slide to the pitch of the slide-note. The slide-time is set by the length of the slide-note.
- **Add a slide note** - Click the **slide switch** and then click as normal to add a note to the Piano roll.
- **De-select the slide mode** - Click the slide button again.
- [Channel Settings > Misc](#) - **Polyphony > SLIDE** (even if grayed out) will interact with Portamento notes, and **Time > GATE** will interact with Portamento and Slide notes.

NOTE: The General settings [Click-and-hold & special gesture functions](#) activate cycle slide/porta. Click and hold when adding a note to cycle through slide & porta event modes.

How slide notes work

Slide notes are used to create '**glissando**' effects. That is, a continuous slide upwards or downwards in pitch between two notes.

- **Notes** - Slide notes **must overlap** the note they slide. A slide note, played on its own, without overlapping a standard note, will not produce sound. The length of the overlap sets the slide time.

When the slide note is reached the pitch of the original (standard) note will start sliding toward the slide notes pitch, reaching the same pitch at the end of the slide note event. The length of the slide note controls the rate of pitch-change of the slide. If the note being slid, ends before the slide notes pitch has been reached, the slide will terminate prematurely.

- **Chords** - If notes are overlapping, so they form part of a chord, the topmost note is used as the reference for the pitch offset (see picture below). All notes in the Chord will slide so the topmost note reaches the same pitch as the slide note, with all notes in the chord moving the same note interval. After the slide event ends, notes still remain offset from their original pitch. **Note colors** can be used to create independent slides for up to 16 notes in a chord. Notes of the same color group, slide each other.

The screenshot displays the 'Piano roll - FLEX' interface. On the left, a piano keyboard is visible with notes B5, A5, G5, F5, E5, D5, C5, B4, A4, G4, F4, and E4. The main area shows two musical staves. The left staff, titled 'Chords slide and maintain note intervals', shows a green slide note (G5) overlapping a chord of four notes (G4, A#4, C#5, G5). All four notes slide together, maintaining their relative intervals. The right staff, titled 'Slide notes independently with note colors', shows a purple slide note (E5) overlapping a chord of four notes (G4, A#4, C#5, G5). The notes are color-coded: G4 and C#5 are green, A#4 is blue, and E5 is purple. Each note slides independently toward the pitch of the slide note it overlaps, with notes of the same color sliding together.

NOTE: Slide events have all the usual note properties (velocity, panning, cutoff and resonance, etc). During slides all note properties move toward those set in the slide note. Slides themselves do NOT produce sound, although they allow preview when created/moved.

How portamento notes work

Portamento applies only to the transition from one note to the next. This is basically a fast slide, only at note transitions. The effect is most audible when larger note intervals are used.

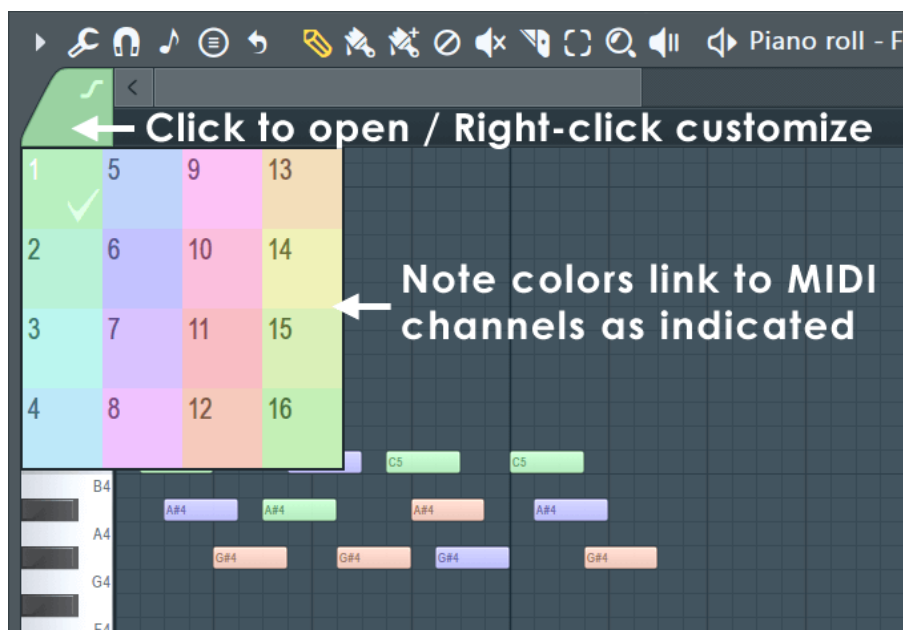
The **Portamento** indicator should be set on the note to receive the portamento effect.

Color Groups (Slides, MIDI, Editing)

Editing - To select the note/slide color, click the Color selector then one of the 16 colors on the **color group selector** and click in the Piano roll to add notes. To **change the color group of existing notes**, make a selection and use the [Piano roll menu > Edit > Change color \(Alt/Opt+C\)](#) option.

By creating color groups you can choose to edit the properties of only notes of the selected color. This can be handy when there are overlapping notes in the same Piano roll that you wish to manipulate independently. Colors are used for independent processing of notes in the Piano roll, these include:

- **MIDI Channels** - Each color is locked to a particular MIDI channel. To see the relationships, hover the mouse over each color and look at the hint bar where the MIDI channel number is displayed. **NOTE:** This only applies to plugins hosted by the [Wrapper](#), in other words, these MIDI channels will only be visible to the plugin associated with the Piano roll. To control external MIDI hardware on separate MIDI channels use the [MIDI Out](#) plugin.



NOTE: You can load and save complete note color palettes from the [Piano roll menu > View > Note colors](#) option.

- **Sliding** - Slides will apply only to notes of the same color as the slide note event. For example, yellow slide note will bend the pitch of yellow notes, but will ignore green notes. This way you can have up to 16 notes sliding simultaneously in different directions.

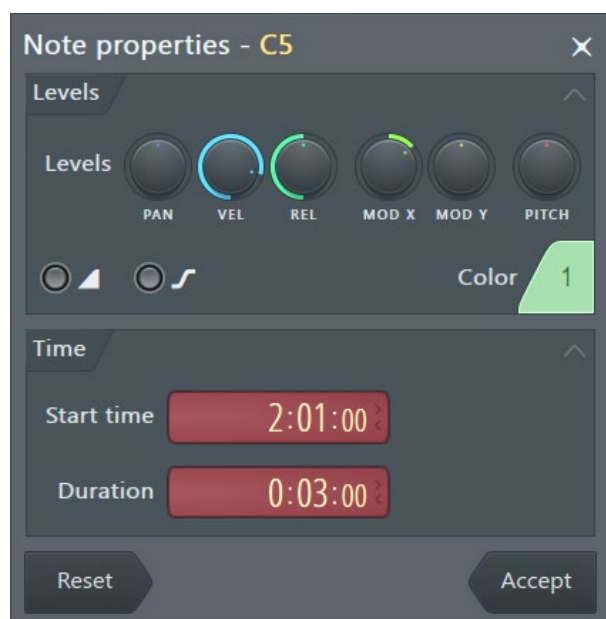
Slide color groups are also used with the **Mono** mode in the polyphony settings (see [Miscellaneous Channel Settings](#)). Mono mode applies to each color group separately. So using all four colors actually can result in a polyphony of up to 4 voices at once.

- **Recording notes** - To record incoming MIDI Channels from a controller or MIDI sequencer to the correct MIDI note colors, **Right-Click** the instrument's Channel Rack button and activate the ['MIDI channel through'](#) option.

NOTE: Some plugins use note colors for special functions. For example [Harmless](#), [Harmor](#) and [Slicex](#).

Note Properties Dialog

The Note Properties pop-up is an alternative way to set notes properties. It's particularly useful when you need to set different properties for notes that start simultaneously (so their properties appear as one in the integrated event editor). Double-click a note to display the properties pop-up box.



- **Levels** - These controls allow changes to **per note** - panning (**PAN**), note on velocity (**VEL**), release velocity (**REL**), channel filter cutoff (**MODX**) and channel filter resonance (**MODY**). **NOTE:** MOD X/Y can also link to custom parameters as set in native plugins. [Harmor](#) has Mod X/Y controls that map to these parameters, for example. Release velocity is supported in Sytrus, Harmor, Harmless and Ogun.
- **Slide**  - Slide ON / Slide OFF. Determines if the note will respond to slide events. **Porta**  - ON, inverts the portamento state for this note. If the global portamento (see [Misc Channel Settings](#)) for this channel is off, for this note it is on and vice versa.
- **Color** - Click on this to select the color channel group the note is on.

NOTE: If the note you double-click is a part of a selection, then the properties you set apply to all notes in that selection. The Time section is not available in that case, because the selected notes might have different length or start point.

- **Time** - Change the **Start Time** (note position) in bar:step:tick. The length of a 'tick' depends on the [PPQ setting](#). Reducing PPQ mid-song will reposition any notes not falling on an exact Bar or Step boundary.
- **Duration** - Change the **Note duration** in bar:step:tick. The length of a 'tick' depends on

the [PPQ setting](#). Reducing PPQ mid-song will reposition any notes not falling on an exact Bar or Step boundary.

- **Reset** - Resets note properties to levels they had before launching the properties box.
- **Accept** - Apply your awesome changes!